

1. UI + Permissions (Who Sees What)

<i>UI Module</i>	<i>Administrator</i>	<i>Doctor</i>	<i>Nurse</i>
<i>Registration</i>	Register users	Register self	Register self
<i>Login</i>	Yes	Yes	Yes
<i>Dashboard</i>	Full access	Clinical view	Referral view
<i>Patient Records</i>	Create, Read, Update, Delete	Read, Update clinical notes	Create, Read, Update
<i>Referral Management</i>	Create, Assign, Update, Delete	Accept, Reject, Complete	Create and Track
<i>Referral Tracking</i>	View all referrals	View assigned referrals	View created referrals
<i>Login Audit</i>	Full access	No	No
<i>User Management</i>	Create/Disable Users	No	No
<i>Chatbot</i>	Yes	Yes	Yes
<i>Reports & Statistics</i>	Full access	Clinical statistics	Referral statistics

Registration Page

Visible fields:

- Username
- Email
- Password
- Confirm Password
- Role Selection
 - Administrator
 - Doctor
 - Nurse

Note : The selected role is stored as the Cognito custom attribute:
custom:role

Dashboard UI

Administrator Dashboard

Widgets:

- Total Users
- Total Patients
- Total Referrals
- Pending Referrals
- Completed Referrals
- Canceled Referrals
- Referrals on hold
- Referrals in progress
- System Statistics

Actions:

- Manage Users
- View Patients
- View Referrals
- View Audit Logs

- System Reports

Doctor Dashboard

Widgets:

- My Referrals
- Accepted Referrals
- Pending Referral
- Completed Referrals

Actions:

- View Patient Records
- Accept Referral
- Reject Referral
- Pending Referral
- Complete Referral

Nurse Dashboard

Widgets:

- Referrals Created
- Pending Referrals
- Canceled Referrals
- Referrals on hold

Actions:

- Register Patient
- Create Referral
- Cancel Referral
- Hold Referral
- Track Referral Status

2. Database Tables

The proposal only explicitly mentions:

- User authentication
- Patient Record Management
- Referral Tracking
- Login Auditing
- AI Chatbot integration
- MySQL on AWS RDS

The actual schema is not defined in the proposal, so the following is the logical database design derived from the documented functionality.

authentication_users

Stores Cognito-linked users.

Field	Type
user_id	BIGINT PK
cognito_sub	VARCHAR(255))

Field	Type
username	VARCHAR(100)
email	VARCHAR(255)
role	VARCHAR(50)
facility	VARCHAR(255)
employee_number	VARCHAR(100)
created_at	DATETIME
updated_at	DATETIME

patients

Stores patient records.

Field	Type
patient_id	BIGINT PK
first_name	VARCHAR(100)
last_name	VARCHAR(100)
date_of_birth	DATE
gender	VARCHAR(20)
contact_number	VARCHAR(20)
address	TEXT
medical_history	TEXT
created_by	BIGINT FK
created_at	DATETIME

Supports:

- Create Patient
- Retrieve Patient
- Patient Record Management Module

referrals

Core referral management table.

Field	Type
referral_id	BIGINT PK
patient_id	BIGINT FK

Field	Type
referring_facility	VARCHAR(255)
receiving_facility	VARCHAR(255)
referral_reason	TEXT
priority	ENUM
status	ENUM
created_by	BIGINT FK
assigned_doctor	BIGINT FK
created_at	DATETIME
updated_at	DATETIME

Status values:

Pending
Accepted
Completed
Rejected

referral_status_history

Required for real-time tracking.

Field	Type
history_id	BIGINT PK
referral_id	BIGINT FK
old_status	VARCHAR(50)
new_status	VARCHAR(50)
changed_by	BIGINT FK
notes	TEXT
changed_at	DATETIME

login_audit

Tracks authentication activity.

Field	Type
audit_id	BIGINT PK
user_id	BIGINT FK
login_time	DATETIME
logout_time	DATETIME

Field	Type
ip_address	VARCHAR(100)
device_info	TEXT
login_status	VARCHAR(50)

chatbot_conversations

Stores chatbot interactions.

Field	Type
conversation_id	BIGINT PK
user_id	BIGINT FK
message	TEXT
response	TEXT
created_at	DATETIME

Supports Amazon Lex integration.

3. User Requirements (All User Perspectives)

User Registration

The user:

1. Opens Registration Page
2. Enters:
 - Username
 - Email
 - Password
 - Role
3. System registers user in Cognito
4. Cognito sends verification code
5. User verifies email
6. User logs in
7. JWT token issued
8. User redirected to Dashboard

This flow is defined in the guide's authentication data flow.

Administrator Requirements

The administrator shall be able to:

- Register and authenticate
- View all patients
- View all referrals

- Track referral status
 - Manage users
 - Assign roles
 - View login audit logs
 - Access reports and statistics
 - Use AI chatbot
 - Monitor system activity
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Nurse Requirements

The nurse shall be able to:

- Register and authenticate
 - Create patient records
 - Search patient records
 - Create referrals
 - View referral status
 - Update referral information
 - Use chatbot for record retrieval and referral queries
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Doctor Requirements

The doctor shall be able to:

- Register and authenticate
 - Access patient records
 - View assigned referrals
 - Accept referrals
 - Reject referrals
 - Complete referrals
 - Update referral notes
 - Use chatbot to retrieve information
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System Requirements

The system shall:

- Provide secure login and authentication
- Support role-based access control
- Store patient records centrally
- Provide referral tracking
- Support referral status updates
- Maintain audit trails
- Provide AI chatbot functionality
- Operate on cloud infrastructure
- Use a centralized MySQL database
- Reduce referral delays
- Improve communication between healthcare providers